Graph Theory-L15 Dual Graph

01 October 2025 18:04

Graph Algorithm and Applications

Course Code: 18B14CI645

Credit: 3

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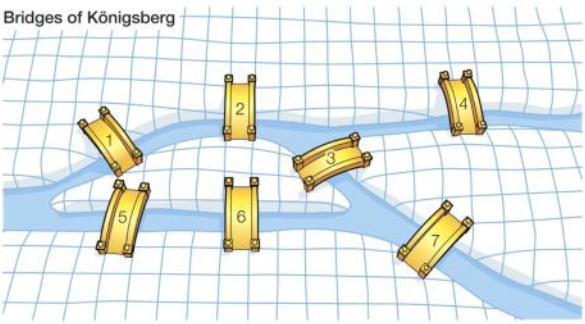
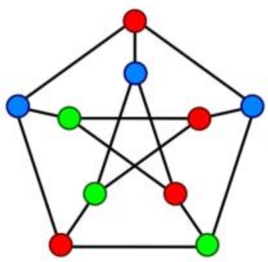
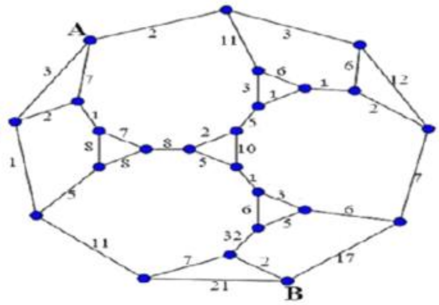






Dual Graphs and Cartesian Product

Dual Graphs

- they are fundamentally related to plane graphs.
- they are often used to create graphs from a graph.

Graph operations.

The Planarity Theorems of MacLane and Whitney 's theorem.
for Graph–like Spaces

Robin Christian, R. Bruce Richter, Brendan Rooney

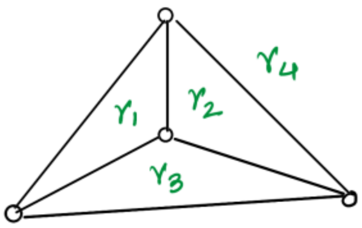
March 12, 2009

- If original graph is G then dual of it will be G^*

Steps to create Dual Graphs

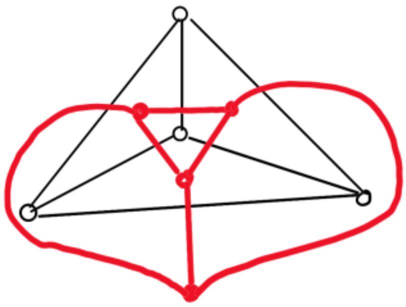
(i) check no. of faces / regions.

$f = 4$



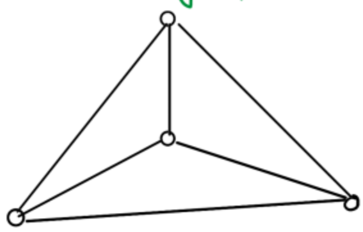
(ii) create one vertex in each face.

(iii) Connect the vertices such that you cross each edge seperating them once.



	r_1	r_2	r_3	r_4
r_1	0	1	1	1
r_2	1	0	1	1
r_3	1	1	0	1
r_4	1	1	1	0

Plane graph



G

Dual Graph

G*

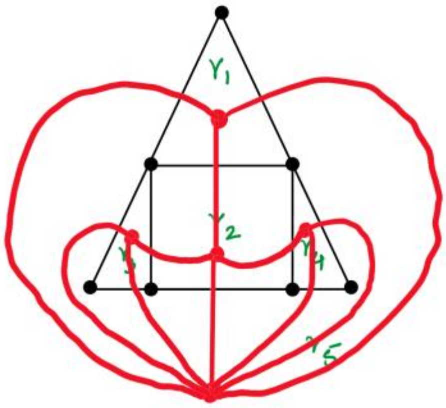
What is the relation?

$n = 4$ $\qquad\qquad$ $n^* = 4$

$m = 6$ $\qquad\qquad$ $m^* = 6$

$f = 4$ $\qquad\qquad$ $f^* = 4$

To understand it well let us solve some problems

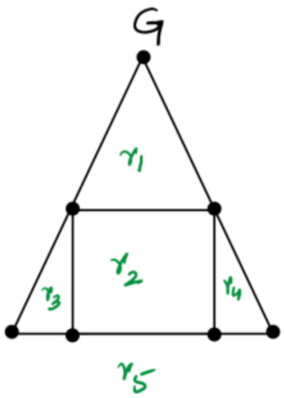


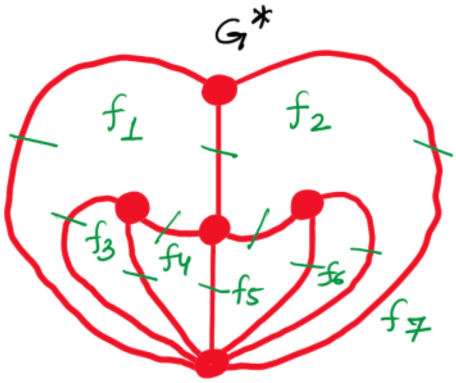
(i) Mark the regions

(ii) create vertices in regions

(iii) create the edges.

	r_1	r_2	r_3	r_4	r_5
r_1	0	1	0	0	2
r_2	1	0	1	1	1
r_3	0	1	0	2	2
r_4	0	1	0	0	2
r_5	2	1	2	2	0

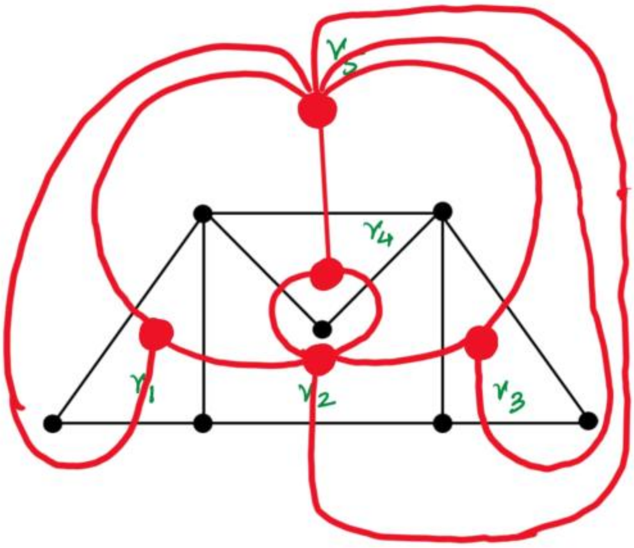




$n = 7$
$m = 10$
$f = 5$

$n^* = 5$
$m^* = 10$
$f^* = 7$

$\therefore \; n = f^*$
$m = m^*$
$f = n^*$

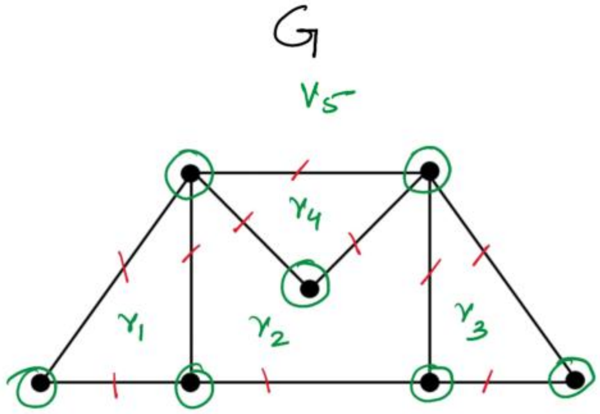


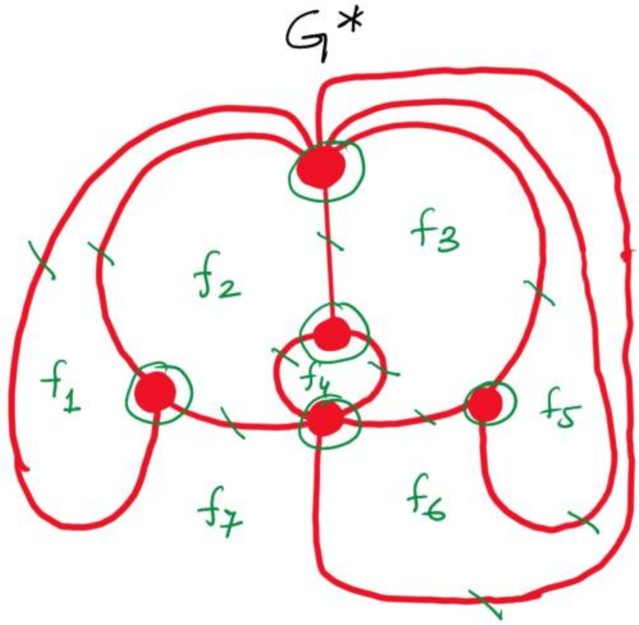
(i) Mark the regions

(ii) create vertices in regions

(iii) create the edges.

	r_1	r_2	r_3	r_4	r_5
r_1	0	1	0	0	2
r_2	1	0	1	2	1
r_3	0	1	0	0	2
r_4	0	2	0	0	1
r_5	2	1	2	1	0



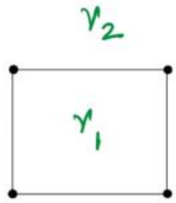


$n = 7 \qquad n^* = 5$

$m = 10 \qquad m^* = 10$

$f = 5 \qquad f^* = 7$

Draw the Dual Graph for the following Plane Graph



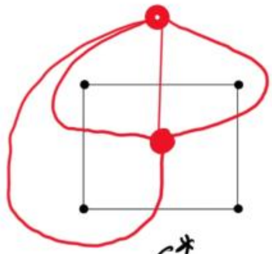
G

$n = 4$

$m = 4$

$f = 2$

	r_1	r_2
r_1	0	4
r_2	4	0

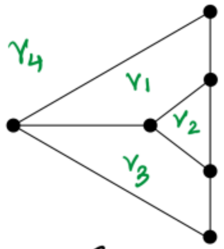


G^*

$n^* = 2$

$m^* = 4$

$f^* = 4$



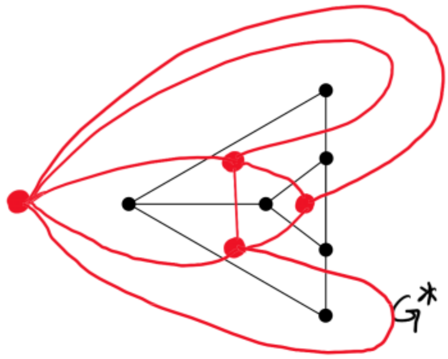
G

$n = 6$

$m = 8$

$f = 4$

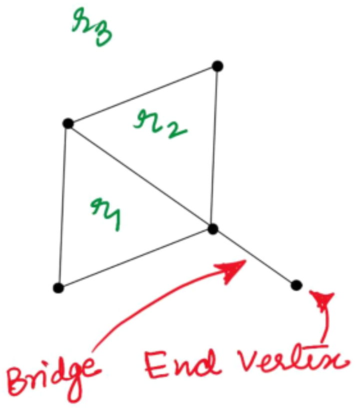
	r_1	r_2	r_3	r_4
r_1	0	1	1	2
r_2	1	0	1	1
r_3	1	1	0	2
r_4	2	1	2	0



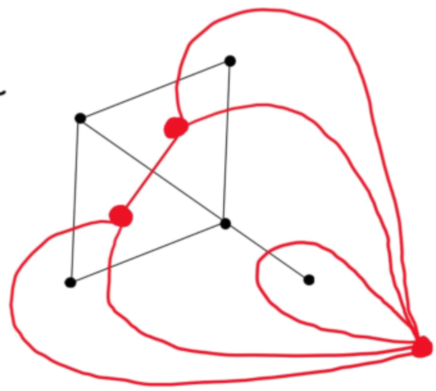
$n^* = 4$

$m^* = 8$

$f^* = 6$

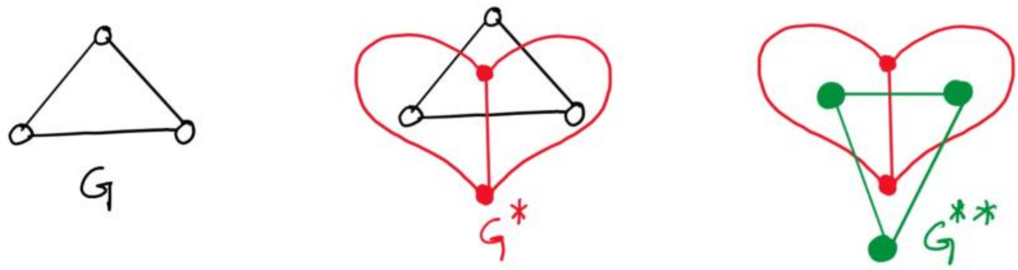


	r_1	r_2	r_3
r_1	0	1	2
r_2	1	0	2
r_3	2	2	0



Note :- when ever there is an end vertex / bride then that always form a loop.

Draw the Dual of the Dual of the following Planar Graph



Observation:-

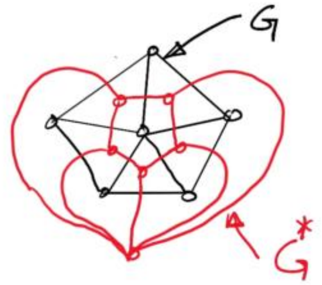
$$G \cong G^{**}$$

Graph(G) for which its dual of dual (G^{**}) is same as G (i.e. $G \cong G^{**}$) is called a self dual graph.

Points to Remember for Dual Graphs

- G and G* have the same number of edges.
- The number of vertices in G* is equal to the number of faces in G and the number of faces in G* is equal to the number of vertices in G.
- An edge forming a self-loop in G becomes a pendant edge in G*.
- A pendent vertex in G becomes self-loop in G*.
- Edges that are in series in G produces parallel edges in G* and parallel edges in G produces edges in series in G*.
- G* is planar.
- If G* is a dual of G then G is dual of G*.

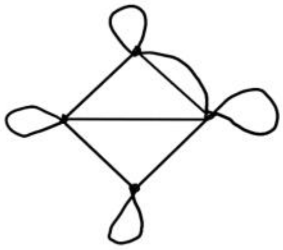
Draw the dual of w_6 and tell it's relation with w_6.

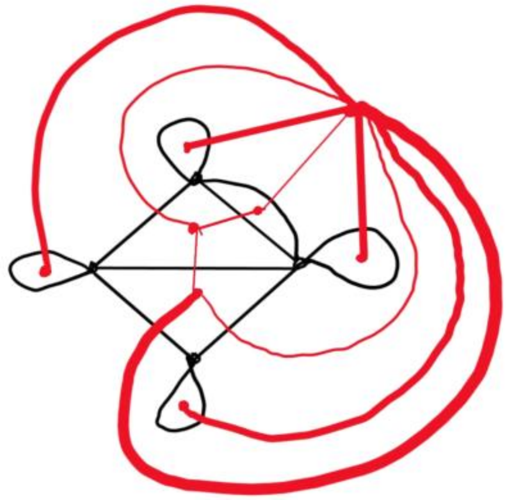


$G \cong G^*$

Evaluate the number of pendants in the dual of the following graph. Then draw its dual.

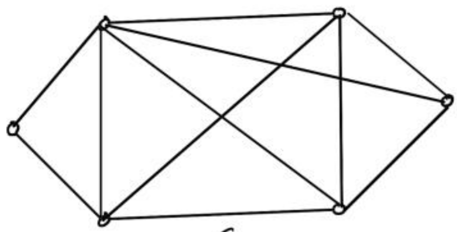
Ans – 4



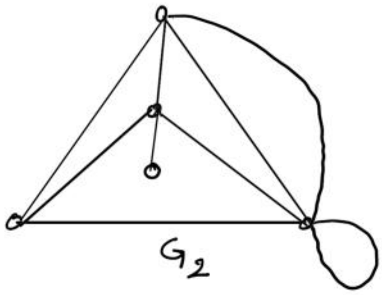


Find the Dual Graph for the following Plane Graph and validate your answer

(Homework problem)



G_1



G_2

Hint :- validation is done using

$m^* - n^* + f^* = 2$ or

$n^* = f$

$m^* = m$

$f^* = n$

Cartesian Product of sets

let A and B are 2 sets such that

$a \in A$ and $b \in B$

$A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$

Cartesian product (or cross)

$A = \{1, 2, 3\}$

$B = \{a, b\}$

$A \times B = \{1a, 2a, 3a, 1b, 2b, 3b\}$

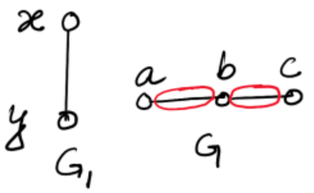
Cartesian Product of Graphs

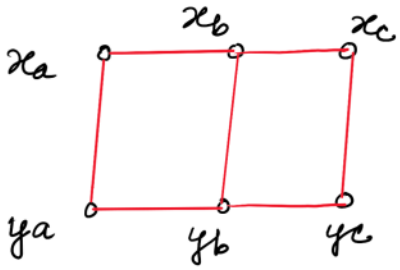
In graphs we use cartesion product on the vertices of 2 graps to create new graphs.

$V_{G_1} = \{x, y\}$

$V_G = \{a, b, c\}$

$V_{G_1} \times V_G = \{xa, xb, xc, ya, yb, yc\}$





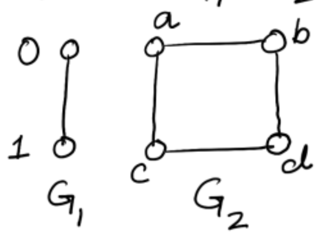
(i) Draw all the vertices

(ii) First hold a, b, and c as constant and draw G_1

(iii) then hold x and y as const. and draw the edges present in G.

Draw $G_1 \times G_2$.

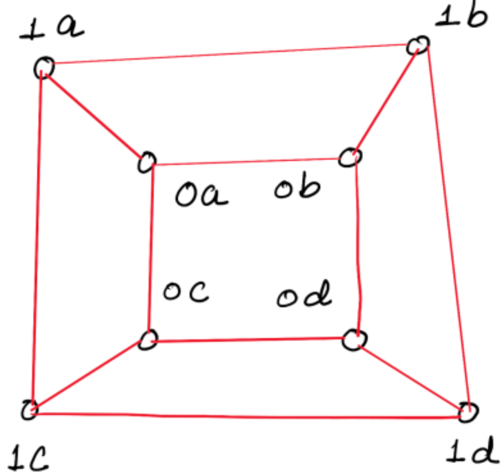
$V_{G_1} = \{0, 1\}$ $V_{G_2} = \{a, b, c, d\}$



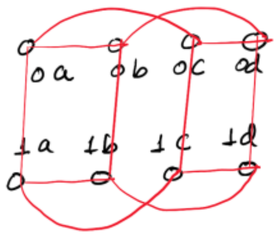
$V_{G_1} \times V_{G_2} = \{0a, 0b, 0c, 0d, 1a, 1b, 1c, 1d\}$

Keep G_2 fixed and draw vertices of G_1

Now draw G_2 edges by keeping G_1 fixed



OR



How many vertices does the graph that is the result of the Cartesian product between a C4 and K5 have?

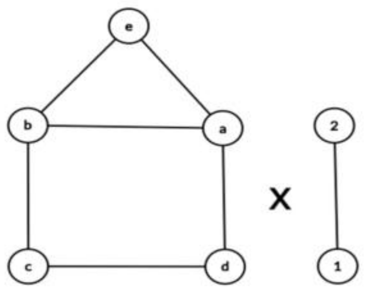
$4 \times 5 = 20$

How many edges does the graph that is the result of the Cartesian product between the below two graphs have? Illustrate if needed.

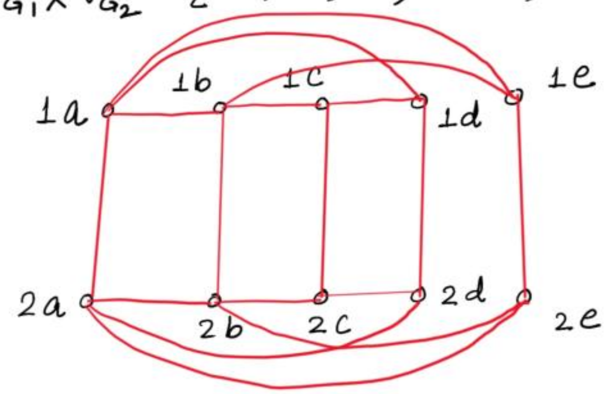
$G = \{a, b, c, d, e\}$

$G_1 = \{1, 2\}$

$V_{G_1} \times V_{G_2} = \{1a, 1b, 1c, 1d, 1e, 2a, 2b, 2c, 2d, 2e\}$



G G_1



Ans:- 17

Draw the Cartesian products of the following graphs (Homework problem)